

Topic 11: Organic chemistry

Premium Answer Booklet

62 questions	2020-2025 Paper 4	Premium readable layout
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Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

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Question 01

13 marks

2020 Feb/March Paper 42 Question 1

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_m20_ms_42.pdf

1(a)	• distillation (1) • fractional (1)
1(b)	• naphtha (1) • bitumen (1)
1(c)(i)	• any two from: • methane ethane propane
1(c)(ii)	• $2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$ • CO_2 and H_2O as products (1) • correct equation (1)
1(c)(iii)	• carbon monoxide
1(d)(i)	• kerosene is more viscous • gasoline is more flammable
1(d)(ii)	• kerosene has molecules with a longer carbon chain
1(e)	• hydrogen + oxygen $\rightarrow$ water

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 02

8 marks

2020 Feb/March Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_m20_ms_42.pdf

5(a)(i)	• monomer
5(a)(ii)	• any hydrocarbon with one C=C bond (with both C atoms having 4 bonds) (1) • structure of methylpropene (1)
5(b)	• water is a product (when polymer is made)
5(c)(i)	• any correct amide link between any two blocks showing all atoms and all bonds (1) • correct orientation of 3 inter-block amide links (1) • continuation bonds (1)
5(c)(ii)	• nylon

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 03

17 marks

2020 May/June Paper 41 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s20_ms_41.pdf

5(a)	• sugar(s) • fermentation • ethene • hydration
5(b)	• (acidified) potassium manganate(VII)
5(c)(i)	• (Liebig) condenser
5(c)(ii)	• arrow at the lower inlet
5(d)(i)	• methanal • C_4H_8O
5(d)(ii)	• $C_nH_{2n}O$
5(e)(i)	• functional group
5(e)(ii)	• 4 x C-H dot cross bonds and 1 C-C dot cross bond • 1 x C=O dot cross bond • non-bonding electrons on O
5(f)(i)	• (structural) isomers
5(f)(ii)	• any structure with correct valencies and formula of C_3H_6O • C=O bond on second carbon (of a chain of 3)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 04

16 marks

2020 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry

Cross-topic links: Topic 5: Chemical energetics, Topic 2: Atoms, elements and compounds, Topic 7: Acids, bases and salts

Answer source label: 0620_s20_ms_42.pdf

6(a)(i)	• substitution
6(a)(ii)	• ultraviolet light / ultraviolet radiation
6(a)(iii)	• No separate point listed.
6(b)(i)	• only one product / double bond becomes single bond / two molecules join (to make one molecule)
6(b)(ii)	• (energy required to break bonds =) 854 (1) • (energy given out when bonds form =) 1025 (1) • overall energy change $854 - 1025 = -171$ (1)
6(c)(i)	• alkene (1) • alcohols / alkanols (1) • carboxylic acids/alkanoic acids (1)
6(c)(ii)	• turns colourless / decolourised (1) • bubbles / fizzing / effervescence (1)
6(d)(i)	• C-C (1) • each C bonded to -CH₂OH / -COOH and H with no other atoms + extension bonds (1)
6(d)(ii)	• polyester • C • C • COOH • CH₂OH • H • H

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 05

10 marks

2020 May/June Paper 43 Question 1

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_s20_ms_43.pdf

1(a)(i)	• D
1(a)(ii)	• C
1(a)(iii)	• C
1(a)(iv)	• B
1(a)(v)	• D AND E or A
1(a)(vi)	• C
1(a)(vii)	• or E
1(a)(viii)	• C • C • C • H • H • H • H • H • H • O • H • H
1(b)	• donate or lose protons • does not fully ionise / partially ionises / forms an equilibrium mixture

Easy explanation / reminder

- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 06

6 marks

2020 May/June Paper 43 Question 8

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s20_ms_43.pdf

8(a)(i)	• No separate point listed.
8(a)(ii)	• No separate point listed.
8(a)(iii)	• water
8(b)	• non-biodegradable / running out of space / toxic / leaching
8(c)	• (complex) carbohydrate / protein

Easy explanation / reminder

- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 07

16 marks

2020 Oct/Nov Paper 41 Question 6

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 3: Stoichiometry, Topic 2: Atoms, elements and compounds
Answer source label: 0620_w20_ms_41.pdf

6(a)(i)	• alkanes
6(a)(ii)	• one mark each for any two of: • same chemical properties • same functional group • same general formula • (consecutive members) differ by CH₂ • common (allow similar) methods of preparation • physical properties vary in predictable manner / show trends / gradually change OR example of a physical property • variation i.e. melting point / boiling point / volatility (1)
6(b)	• 3.01×10^{23} (molecules)
6(c)(i)	• HCl
6(c)(ii)	• No separate point listed.
6(c)(ii)	• substitution
6(d)(i)	• from: orange (1) • to: colourless (1)
6(d)(ii)	• contains no double bonds/ethane only contains single bonds
6(e)(i)	• (1) • (1)
6(e)(ii)	• C₂H₄
6(f)(i)	• (C = 85.7, H = 14.3, Mr 112) • C = 85.7 • $12 = 7.14$ H = 14.3 • = 14.3 (1) • (ratio = 7.13 : 14.3 = 1 : 2) • CH₂ (2)
6(f)(ii)	• C₈H₁₆

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 08

8 marks

2020 Oct/Nov Paper 41 Question 7

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w20_ms_41.pdf

7(a)(i)	• sugar or $C_6H_{12}O_6$, is renewable / sustainable
7(a)(ii)	• slow(er) process
7(b)(i)	• ethanoic acid
7(b)(ii)	• oxidation
7(c)(i)	• ethyl methanoate
7(c)(ii)	• No separate point listed.
7(d)	• ethanol: (forces of attraction) between molecules (1) • sodium chloride: (force of attraction) between positive and negative ions/ionic bonding (1)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 09

10 marks

2020 Oct/Nov Paper 42 Question 4

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w20_ms_42.pdf

4(a)	• hydrocarbon
4(b)(i)	• (only) one product is formed
4(b)(ii)	• 3 single bonded C atoms • entire molecule correct
4(b)(iii)	• orange (1) to colourless (1)
4(b)	• (iv • structure of but-1-ene • structure of but-2-ene
4(c)	• $\text{C}_3\text{H}_8 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{HCl}$ • HCl as product • rest of equation

Easy explanation / reminder

- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 10

22 marks

2020 Oct/Nov Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_w20_ms_42.pdf

5(a)(i)	• presence of an acid
5(a)(ii)	• oxidation
5(a)(iii)	• all single bonding dot and cross pairs correct (1) • double C=O bond dot and cross pairs are correct (1) • Complete diagram is correct (1)
5(b)	• partial (dissociation) (1) • full / 100% (dissociation) (1) • both acid colours (1) • HCl indicating a lower pH acid colour than CH₃COOH (1) • fizzing OR dissolving / disappearing in both (1) • either observation happens quicker with HCl (1)
5(c)(i)	• water
5(c)(ii)	• acid (catalyst)
5(c)(iii)	• any one alcohol group on a molecule (1) • structure of propan-2-ol (1)
5(d)(i)	• the atoms / group of atoms which give (any molecule its) chemical properties
5(d)(ii)	• C₃H₅O₂
5(d)(iii)	• Mr of HO-CH₂-CH₂-OH = 62 (1) • %ge = $100 \times 32/62 = 52\%$ (1)
5(d)(iv)	• any correct ester link between any two blocks showing all atoms and all bonds (1) • correct orientation of 3 inter-block ester links and continuation bonds (1)
5(d)(v)	• Terylene

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 11

14 marks

2020 Oct/Nov Paper 43 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w20_ms_43.pdf

6(a)(i)	• compounds with the same molecular formula (1) • different structural formula (1)
6(a)(ii)	• diagram (1) • diagram (1) • ethanoic acid (1) • methyl methanoate (1)
6(b)(i)	• No separate point listed.
6(b)(ii)	• condensation
6(b)(iii)	• 1 for each correct functional group
6(c)	• one correct linkage fully displayed (1) • the whole structure fully correct (1)
6(d)(i)	• proteins
6(d)(ii)	• amino acids

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 12

20 marks

2021 Feb/March Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_m21_ms_42.pdf

5(a)	• R and U only
5(b)	• aqueous bromine (1) • turns colourless / decolourises (1)
5(c)(i)	• but-2-ene
5(c)(ii)	• structure of methyl propene
5(d)(i)	• C₁₂H₂₆
5(d)(ii)	• P, R and U
5(e)	• steam (1) • catalyst (1) • One other condition (1): • either 60 atm or 300 °C
5(f)(i)	• oxidation
5(f)(ii)	• potassium manganate(VII)
5(g)(i)	• ester
5(g)(ii)	• name = methanol • structure = • name = butanoic acid structure =
5(g)(iii)	• C₅H₁₀O₂

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 13

9 marks

2021 Feb/March Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_m21_ms_42.pdf

6(a)	• monomers
6(b)(i)	• any hydrocarbon with one C=C bond (with both C atoms having 4 bonds) (1) • structure of but-2-ene (1)
6(b)(ii)	• addition
6(c)(i)	• protein
6(c)(ii)	• any correct amide link between any two blocks showing all atoms and all bonds (1) • identical orientation of both inter-block amide links including terminal groups with correct orientation (if shown) (1) • continuation bonds on polymer (1)
6(c)(iii)	• hydrolysis (1)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 14

15 marks

2021 May/June Paper 41 Question 7

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_s21_ms_41.pdf

7(a)	• 48.65 / 12 8.11 / 1 43.24 / 16 (1) • OR evaluation • 4.05 8.11 2.7(0) • divide all by smallest • OR 1.5 : 3 : 1 • OR 6 : 3 : 2 (1) • C₃H₆O₂ (1) ALLOW symbols in any order
7(b)	• (Mr of CH₄O = 32) • CH₄O (1)
7(c)(i)	• C_nH_{2n}O₂ • OR • C_nH_{2n+1} COOH
7(c)(ii)	• butanoic acid (1) • fully displayed carboxylic acid group (1) • correct structure of butanoic acid showing all atoms and bonds (1)
7(c)(iii)	• homologous series
7(d)(i)	• brown to colourless
7(d)(ii)	• C₉H₂₀ (1) • 2C₃H₆ (1)
7(d)(iii)	• addition
7(d)(iv)	• any one repeat unit (1) • both repeat units fully correct (1)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 15

12 marks

2021 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s21_ms_42.pdf

6(a)(i)	• carboxylic acid
6(a)(ii)	• any correct displayed ester link between any two blocks showing all atoms and all bonds (1) • correct orientation of three displayed inter-block ester links with correct orientation (1) • continuation bonds on polyester (1)
6(a)(iii)	• water
6(b)(i)	• two blocks linked by -O-
6(b)(ii)	• acid (and heat) (1) • enzymes (ignore names) (1)
6(b)(iii)	• chromatography
6(c)(i)	• fermentation
6(c)(ii)	• $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ • C_2H_5OH (1) • rest of equation (1)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 16

7 marks

2021 May/June Paper 43 Question 1

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 4: Electrochemistry
Answer source label: 0620_s21_ms_43.pdf

1(a)	• polymerisation
1(b)	• fractional distillation
1(c)	• fermentation
1(d)	• distillation
1(e)	• electrolysis
1(f)	• chromatography
1(g)	• filtration • Question Answer

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 17

15 marks

2021 May/June Paper 43 Question 7

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_s21_ms_43.pdf

7(a)	• 11.63 / 1 69.77 / 12 18.6 /16 • Or evaluation • 11.63 5.81 1.16 • Or ratio • 10 : 5 : 1 (1) • C₅H₁₀O (1)
7(b)	• (Mr of CH₂O =) 30 (1) • C₂H₄O₂ (1)
7(c)(i)	• structural isomers
7(c)(ii)	• carboxylic acids • Question Answer
7(c)(iii)	• name of ester, corresponding to C₃H₆O₂: ethyl methanoate or methyl ethanoate (1) • correctly displayed ester linkage (1) • fully correct displayed formula corresponding to C₃H₆O₂ and matching named ester (1)
7(c)(iv)	• displayed formula of propanoic acid (1) • propanoic acid (1)
7(d)	• Step 1: • cracking (1) • C₁₀H₂₂ -> C₂H₄ + C₈H₁₈ (1) • Step 2: • (catalytic) addition (1) • C₂H₄ + H₂O -> C₂H₅OH (1)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 18

15 marks

2021 Oct/Nov Paper 41 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w21_ms_41.pdf

5(a)	homologous series
5(b)	C_nH_{2n}
5(c)(i)	orange to colourless
5(c)(ii)	(only) one product is formed
5(c)(iii)	$C_4H_8 + Br_2 \rightarrow C_4H_8Br_2$ $C_4H_8Br_2$ (1) - equation fully correct (1)
5(c)(iv)	(poly) but-2-ene
5(c)(v)	- but-1-ene (1) - structure of but-1-ene (1)
5(d)(i)	C_2H_4O
5(d)(ii)	$2C_3H_7COOH + Na_2CO_3 \rightarrow 2C_3H_7COONa + H_2O + CO_2$ C_3H_7COONa (1) - equation fully correct (1)
5(d)(iii)	- methyl butanoate (1) - ester link (1) - rest of structure (1)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 19

12 marks

2021 Oct/Nov Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w21_ms_42.pdf

6(a)	• $C_nH_{2n+1}OH$ (1)
6(b)	• not all atoms are C or H
6(c)(i)	• C_2H_4
6(c)(ii)	• glucose
6(d)	• $C_2H_5OH + 3O_2 \rightarrow 3H_2O + 2CO_2$ • species (1) correct equation (1)
6(e)(i)	• $C_3H_5O_2$
6(e)(ii)	• any one ester link (1) • rest of molecule (1)
6(e)(iii)	• water
6(f)(i)	• carboxylic acid
6(f)(ii)	• structure of ethanedioic acid

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 20

16 marks

2021 Oct/Nov Paper 43 Question 7

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w21_ms_43.pdf

7(a)	• 1 for each for any 2 of: • (same) general formula • similar chemical properties • (consecutive members) differ by CH₂ • same functional group • common (allow similar or same) methods of preparation • physical properties and vary in predictable manner or show trends or gradually change • OR example of a physical property variation, e.g. • melting points increase • boiling points increase • volatility decreases
7(b)(i)	• cracking
7(b)(ii)	• $\rightarrow \text{C}_2\text{H}_4 + \text{C}_4\text{H}_{10}$ (2)
7(b)(iii)	• $\text{C}_2\text{H}_4 + \text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_5\text{OH}$
7(b)(iv)	• hydration / addition
7(b)(v)	• 1 mark for ANY 1 of: • 300 °C • 60 atmospheres • H₃PO₄
7(c)	• 1 mark each for any 2 of: • carbohydrates are renewable • fossil fuels are non-renewable OR fossil fuels conserved • lower temperature OR lower energy • hydration is equilibrium meaning lower yield • lower pressure (used)
7(d)(i)	• substitution
7(d)(ii)	• addition
7(d)(iii)	• any one repeat unit with extension bonds (1) • both units fully correct (1)
7(e)	• 1 mark for each functional group fully correct (2)

Easy explanation / reminder

- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 21

11 marks

2022 Feb/March Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_m22_ms_42.pdf

5(a)(i)	• cracking
5(a)(ii)	• $\text{C}_{10}\text{H}_{22} \rightarrow 4\text{C}_2\text{H}_4 + \text{C}_2\text{H}_6$ • $\text{C}_{10}\text{H}_{22}$ as only reactant • formulae of ethene and ethane as only products • correct equation
5(b)(i)	• hydrogen chloride
5(b)(ii)	• ultraviolet light
5(c)(i)	• (only) one product is formed
5(c)(ii)	• Bond energy in breaking bonds • $= [(4 \times 410) + 610 + 240] = 2490 \text{ (kJ / mol)}$ • Use of total E change to find bond energy of $\text{C}_2\text{H}_4\text{Cl}_2$ • $= + 180 = 2490 + 180 = 2670 \text{ (kJ / mol)}$ • Determination of total C-Cl bond energy • $= - [(4 \times 410) + 350] = 2670 - 1990 = 680 \text{ (kJ / mol)}$ • Determination of each C-Cl bond energy • $= / 2 = 680 / 2 = 340 \text{ (kJ / mol)}$

Easy explanation / reminder

- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.
- Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.

END OF ANSWER

Question 22

17 marks

2022 Feb/March Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_m22_ms_42.pdf

6(a)(i)	homologous series
6(a)(ii)	Same functional group
6(a)(iii)	general (formula)
6(a)(iv)	February/March 2022
6(b)	- fractional - distillation
6(c)(i)	- propanoic acid - structure of propanoic acid
6(c)(ii)	methanol
6(d)(i)	structural isomers
6(d)(ii)	- ethyl butanoate - butyl ethanoate
6(e)(i)	- any correct displayed ester link between any two blocks showing all atoms and all bonds - correct orientation of three displayed inter-block ester links with correct orientation - continuation bonds on polyester
6(e)(ii)	condensation
6(e)(iii)	terylene

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 23

19 marks

2022 May/June Paper 41 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s22_ms_41.pdf

5(a)	- the rate of forward reaction equals the rate of the reverse reaction (1) - concentrations of reactants and products are constant (1)
5(b)	- reaction too slow (1) - yield of ethanoic acid too low (1)
5(c)	- increases (1) - decreases (1) - decreases (1)
5(d)	- cobalt (1) - transition element (1)
5(e)	$C_nH_{2n+1}COOH$
5(f)	- COOH fully displayed (1) - whole molecule completely correct (1)
5(g)(i)	methyl butanoate
5(g)(ii)	- butanoic acid (1) - methanol (1)
5(h)	- C 48.65 / 12 - H 8.11 / 1 - O 43.24 / 16 - OR 4.05:8.11:2.70 (1) - fractions shown dividing all by smallest - OR 1.5:3:1 - OR 3:6:2 (1) - C₃H₆O₂ (1)
5(i)	C₄H₈O₂

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 24

15 marks

2022 May/June Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s22_ms_42.pdf

5(a)	• fermentation
5(b)	• (complete) combustion • $\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 3\text{H}_2\text{O} + 2\text{CO}_2$ • reactant species • fully correct equation
5(c)(i)	• ethene
5(c)(ii)	• addition
5(c)(iii)	• steam • catalyst + 60 atm or catalyst + 300 °C
5(d)(i)	• hydrogen • catalyst • high temperature
5(d)(ii)	• $\text{C}_n\text{H}_{2n+2}$
5(e)(i)	• potassium manganate(VII)
5(e)(ii)	• carboxylic acids
5(e)(iii)	• fully displayed formula of ethanoic acid

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 25

9 marks

2022 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s22_ms_42.pdf

6(a)(i)	• No separate point listed.
6(a)(ii)	• diagram of dioic acid showing two fully displayed carboxylic acid groups on unshaded block • diagram of diol showing two fully displayed hydroxyl groups on shaded block
6(a)(iii)	• polyester
6(b)	• one
6(c)(i)	• any hydrocarbon with one C=C bond (with both C atoms having 4 bonds) (1) • structure of but-2-ene (1) • but-2-ene
6(c)(ii)	• cracking

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 26

18 marks

2022 May/June Paper 43 Question 4

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s22_ms_43.pdf

4(a)	• (a substance which) increases the rate of a reaction • remains unchanged or unaffected or without changing mass
4(b)	• increases (1) • decreases (1) • decreases (1) • decreases (1)
4(c)(i)	• $C_nH_{2n+1}OH$
4(c)(ii)	• any 2 from: • $\blacksquare$ • same or similar chemical properties or reactions • $\blacksquare$ • (contain) the same functional group • $\blacksquare$ • (consecutive members) differ by CH_2 • $\blacksquare$ • physical properties vary in predictable manner / show trends / gradually change
4(c)(iii)	• (1) • propan-2-ol (1)
4(d)(i)	• ethyl butanoate
4(d)(ii)	• ethanol (1) • butanoic acid (1)
4(e)	• C 58.82 / 12 H 9.80 / 1 O 31.37 / 16 • OR • 4.90:9.80:1.96 (1) • 2.5:5:1 • OR • 5:10:2 (1) • $C_5H_{10}O_2$(1)
4(f)	• $C_6H_{12}O_2$

Easy explanation / reminder

- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.

END OF ANSWER

Question 27

9 marks

2022 Oct/Nov Paper 41 Question 5

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 5: Chemical energetics
Answer source label: 0620_w22_ms_41.pdf

5(a)	• C_nH_{2n}
5(b)	• orange to colourless
5(c)	• addition
5(d)(i)	• horizontal line below energy level to right hand side of reactants line and labelled $C_2H_4Br_2$ (1) • activation energy 'hump' with upward arrow labelled from the reactants level (1) • one downward arrow starting from the energy level of the reactants and finishing at the energy level of the products (1)
5(d)(ii)	• energy needed to break bonds • $4 \rightarrow C-H + C=C + Br-Br = 4 \rightarrow 410 + 610 + 190 = 2440 \text{ (kJ)} \text{ (1)}$ • energy released in making bonds • $4 \rightarrow C-H + C-C + 2 \rightarrow C-Br = 4 \rightarrow 410 + 350 + 2 \rightarrow 290 = 2570 \text{ (kJ)} \text{ (1)}$ • energy change - = - 130 (kJ / mol) (1)

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 28

12 marks

2022 Oct/Nov Paper 41 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w22_ms_41.pdf

6(a)(i)	• methyl methanoate
6(a)(ii)	• CH₂O
6(b)	• all single bonding dot and cross pairs correct (1) • double C=O bond dot and cross pairs are correct (1) • complete diagram is correct (1)
6(c)	• methanoic acid (1) • structure of methanoic acid (1) • methanol (1) • structure of methanol (1)
6(d)(i)	• same (molecular) formula (1) • different structural formula (1)
6(d)(ii)	• ethanoic acid

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 29

10 marks

2022 Oct/Nov Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 5: Chemical energetics
Answer source label: 0620_w22_ms_42.pdf

5(a)	• C_nH_{2n+2}
5(b)	• needs ultraviolet (light)
5(c)	• substitution
5(d)	• hydrogen chloride
5(e)(i)	• activation energy
5(e)(ii)	• energy (change) of reaction
5(e)(iii)	• energy of products is lower than energy of reactants
5(f)	• energy needed to break bonds • $6 \rightarrow C-H + C-C + Cl-Cl = 6 \rightarrow 410 + 350 + 240 = 3050 \text{ (kJ / mol) (1)}$ • energy released in making bonds • $5 \rightarrow C-H + C-C + C-Cl + H-Cl = 5 \rightarrow 410 + 350 + 340 + 430 = 3170 \text{ (kJ / mol) (1)}$ • energy change - = $-120 \text{ (kJ / mol) (1)}$

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 30

12 marks

2022 Oct/Nov Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w22_ms_42.pdf

6(a)	• $C_nH_{2n+1}COOH$
6(b)	• methanoic acid
6(c)	• CH_2O_2
6(d)	• all single bonding dot and cross pairs correct (1) • double $C=O$ bond dot and cross pairs are correct (1) • four non-bonding cross electrons on each O atom (1)
6(e)(i)	• propyl methanoate (1) • ester linkage with all atoms and all bonds (1) • correct structure of propyl methanoate (1)
6(e)(ii)	• water
6(e)(iii)	• methyl propanoate (1) • butanoic acid (1)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 31

18 marks

2022 Oct/Nov Paper 43 Question 6

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 5: Chemical energetics
Answer source label: 0620_w22_ms_43.pdf

6(a)(i)	• 2670 (1) • 2610 (1) • (+) 60 (1)
6(a)(ii)	• endothermic AND energy released when bonds form is less than energy absorbed to break bonds • OR • endothermic AND overall energy change has a positive sign
6(b)(i)	• No separate point listed.
6(b)(ii)	• but-1-ene (1) • C=C (both C atoms should have 4 bonds) (1) • Displayed formula of but-1-ene (1)
6(b)(iii)	• CH₂
6(c)(i)	• hydrolysis
6(c)(ii)	• acids (1) • enzymes (1)
6(c)(iii)	• chromatography (1) • locating agent (1) • R_f (1)
6(d)(i)	• nylon
6(d)(ii)	• carboxylic acid (1) • amine (1)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 32

17 marks

2023 Feb/March Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_m23_ms_42.pdf

5(a)	• same functional group
5(b)	• carboxylic acids
5(c)	• $C_nH_{2n+1}COOH$
5(d)	• propan-2-ol (1) • displayed formula of propan-2-ol (1)
5(e)(i)	• $C_{10}H_{22} \rightarrow C_3H_8 + C_3H_6 + C_4H_8$ • $C_3H_8 + C_3H_6$ (1) • rest of equation correct (1) • February/March 2023
5(e)(ii)	• cracking
5(f)(i)	• six C atoms joined by single bonds in a chain and with continuation bonds (1) • 3 $\rightarrow$ CH₃ at two C intervals and whole structure correctly displayed (1)
5(f)(ii)	• addition
5(g)(i)	• sodium propanoate
5(g)(ii)	• $CH_3CH_2COO^-$
5(h)(i)	• acid • or • any named inorganic acid, e.g. phosphoric acid
5(h)(ii)	• ethyl propanoate
5(h)(iii)	• displayed formula of any ester linkage (1) • correct structure of ethyl propanoate (1)

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 33

6 marks

2023 May/June Paper 41 Question 1

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_s23_ms_41.pdf

1(a)	• B
1(b)	• A
1(c)	• H
1(d)	• J
1(e)	• G
1(f)	• D

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 34

12 marks

2023 May/June Paper 41 Question 7

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s23_ms_41.pdf

7(a)(i)	• needs or uses ultra violet light
7(a)(ii)	• displayed formula of 1-chlorobutane(1) • displayed formula of 2-chlorobutane(1)
7(b)(i)	• C₄H₆O₃
7(b)(ii)	• alkene(1) • alcohol(1) • carboxylic acid(1)
7(b)(iii)	• turns colourless(1) • bubbles / fizzing / effervescence(1)
7(b)(iv)	• only two carbon atoms joined by a single bond and two additional bonds on each(1) • (1)
7(b)(v)	• polyester

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 35

11 marks

2023 May/June Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 2: Atoms, elements and compounds, Topic 3: Stoichiometry
Answer source label: 0620_s23_ms_42.pdf

5(a)(i)	• substitution
5(a)(ii)	• provide activation energy
5(a)(iii)	• photochemical
5(a)(iv)	• $\text{C}_3\text{H}_8 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{HCl}$ • HCl equation correct
5(b)(i)	• it has a carbon-carbon bond which is not a single bond
5(b)(ii)	• $\text{CH}_3\text{CHClCH}_2\text{Cl}$
5(c)	• displayed formula of propan-1-ol • displayed formula of propan-2-ol • propan-1-ol • (as either name) • propan-2-ol (under displayed formula of propan-2-ol)

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 36

7 marks

2023 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s23_ms_42.pdf

6(a)	• ethyl butanoate
6(b)	• water
6(c)	• C ₃ H ₆ O
6(d)(i)	• 1 repeat unit circled
6(d)(ii)	• displayed diol on correct box (unshaded) • displayed dioic acid on correct box (shaded)
6(d)(iii)	• condensation

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 37

6 marks

2023 May/June Paper 43 Question 1

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s23_ms_43.pdf

1(a)	• J
1(b)	• D
1(c)	• B
1(d)	• G
1(e)	• F
1(f)	• C

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 38

12 marks

2023 May/June Paper 43 Question 7

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s23_ms_43.pdf

7(a)(i)	ultraviolet (light)
7(a)(ii)	- displayed formula of 1-chloropropane - displayed formula of 2-chloropropane
7(b)(i)	- any 2 from: (same) general formula (same / similar) chemical properties or reactions (contain the same) functional group
7(b)(ii)	- single bond between the two C atoms (and nothing on continuation bonds) 1 CH₃ group and 1 H on first carbon and 1 CH₃ group and 1 H on second carbon and whole structure correct
7(c)(i)	- structural formula of propan-1,3-dioic acid - structural formula of ethan-1,2-diol
7(c)(ii)	polyester
7(c)(iii)	- carboxylic acids - alcohols

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 39

17 marks

2023 Oct/Nov Paper 41 Question 6

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_w23_ms_41.pdf

6(a)	• yeast (1) • absence of air (1)
6(b)(i)	• 300 (1) • 6000 (1)
6(b)(ii)	• ethene
6(b)(iii)	• only one product
6(c)(i)	• proton donor
6(c)(ii)	• partial dissociation
6(c)(iii)	• 4 $\rightarrow$ -2 or -8 (1) • $P + (4 \rightarrow -2) = -3 \therefore P = +5$ (1)
6(d)	• uses renewable resources (1) • high rate of reaction (1)
6(e)(i)	• potassium manganate(VII)
6(e)(ii)	• reducing agent
6(f)(i)	• calcium ethanoate
6(f)(ii)	• $(\text{CH}_3\text{COO})_2\text{Ca}$
6(f)(iii)	• hydrogen

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 40

11 marks

2023 Oct/Nov Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_w23_ms_42.pdf

6(a)	• (same) general formula (1) • (contain the same) functional group (1)
6(b)	• -CH₂- unit
6(c)(i)	• alkenes (1) • alkanes (1) • carboxylic acids (1)
6(c)(ii)	• propanoic acid (1) • displayed formula of propanoic acid (1)
6(d)(i)	• working to give the mass of atoms shown in Fig 6.1 • $103 - 14 - (2 \rightarrow 12) - (2 \rightarrow 16) - (4 \rightarrow 1)$ • (= 29) • 'R' = C₂H₅
6(d)(ii)	• proteins

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 41

12 marks

2023 Oct/Nov Paper 43 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w23_ms_43.pdf

6(a)	• any two • general formula • functional group • chemical properties
6(b)	• ethyl propanoate
6(c)(i)	• propan-1-ol (1) • methanoic acid (1)
6(c)(ii)	• C₄H₈O₂
6(d)	• C₂H₄O
6(e)(i)	• condensation
6(e)(ii)	• 2 -COOH groups fully displayed (1) • 2 -NH₂ groups fully displayed (1)
6(f)(i)	• proteins
6(f)(ii)	• amino acids

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 42

12 marks

2024 Feb/March Paper 42 Question 4

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 6: Chemical reactions, Topic 7: Acids, bases and salts
Answer source label: 0620_m24_ms_42.pdf

4(a)(i)	• structural
4(a)(ii)	• CH₂O₂
4(b)	• ethyl methanoate • ester link • correct displayed formula of ethyl methanoate
4(c)(i)	• nothing can enter or leave
4(c)(ii)	• the rate of forward reaction equals (the rate of the) reverse reaction • concentrations of reactants and products are constant • February/March 2024
4(c)(iii)	• increases • decreases • decreases • no effect

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 43

15 marks

2024 Feb/March Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 2: Atoms, elements and compounds, Topic 3: Stoichiometry
Answer source label: 0620_m24_ms_42.pdf

5(a)	• (but-1-ene) has a lower relative molecular mass
5(b)	• carbon dioxide AND water
5(c)(i)	• photochemical
5(c)(ii)	• to provide the activation energy
5(c)(iii)	• substitution
5(c)(iv)	• No separate point listed.
5(d)(i)	• acid
5(d)(ii)	• displayed formula of butan-1-ol • displayed formula of butan-2-ol • butan-1-ol • butan-2-ol
5(e)(i)	• addition (polymerisation) • February/March 2024
5(e)(ii)	• chain of six C atoms joined by single bonds in a chain • three correctly placed C₂H₅ groups • correct structure AND continuation bonds

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 44

13 marks

2024 May/June Paper 41 Question 7

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s24_ms_41.pdf

7(a)(i)	• C 82.76 / 12: H 17.24/1 (1) • OR evaluation i.e. • 6.90:17.24 • fractions showing division of both by smaller ie 6.9 / 6.9 and 17.24 / 6.9 • OR evaluation ie 1:2.5 OR 4:10 • C₂H₅ (1)
7(a)(ii)	• C₅H₁₀
7(b)	• any C=C with both carbons having a valency of 4(1) • correct displayed formula of (cis or trans) but-2-ene(1) • but-2-ene
7(c)	• C₈H₁₆
7(d)(i)	• structural isomers
7(d)(ii)	• ethanoic acid(1) • methanol(1)
7(d)(iii)	• Displayed formula of propanoic acid showing ALL atoms and bonds • (1) • propanoic acid(1)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 45

14 marks

2024 May/June Paper 42 Question 4

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s24_ms_42.pdf

4(a)	• C₂H₄O
4(b)	• methyl propanoate
4(c)	• displayed formula of propanoic acid • propanoic acid
4(d)	• structure of CH₃OH • methanol
4(e)	• decreases • decreases • decreases • no effect
4(f)(i)	• 7 • 3 ■ pH ■ 7
4(f)(ii)	• H⁺(aq)
4(f)(iii)	• universal indicator

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 46

13 marks

2024 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s24_ms_42.pdf

6(a)(i)	• photosynthesis
6(a)(ii)	• $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
6(a)(iii)	• energy from light • presence of chlorophyll
6(b)(i)	• fermentation
6(b)(ii)	• carbon dioxide
6(c)(i)	• 300 °C • 60 atm or 6000 kPa • acid
6(c)(ii)	• only one product is formed
6(c)(iii)	• C-C bond as dot and cross and C-O as 'cross and dot' • all C-H bonds and O-H bond as dot and cross • 4 non-bonding dot electrons on O atom and no other non-bonding electrons on C or H

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 47

14 marks

2024 May/June Paper 43 Question 7

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s24_ms_43.pdf

7(a)(i)	• C 83.33 / 12: H 16.67 / 1 • or • 6.94:16.67 • C 6.94 / 6.94: H 16.67 / 6.94 • or 1:2.4 • or 5:12 • C₅H₁₂
7(a)(ii)	• Mr of C₂H₅ = 29 • 58 / 29 = 2 and C₄H₁₀
7(b)(i)	• C=C
7(b)(ii)	• C • C • any C=C with both carbons having a valency of 4 • displayed formula of • but-1-ene • but-1-ene
7(c)	• 2C₂H₄ • C₈H₁₈
7(d)(i)	• C_nH_{2n} + 10H
7(d)(ii)	• displayed formula of propan-1-ol or propan-2-ol • propan-1-ol • or • propan-2-ol

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 48

17 marks

2024 Oct/Nov Paper 41 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w24_ms_41.pdf

5(a)(i)	• oxygen
5(a)(ii)	• advantage: any one of: • water is the only product • no carbon dioxide produced • more efficient • disadvantage: any one of: • hydrogen needs to be stored at high pressure • hydrogen hard to store • heavy tanks needed to store hydrogen • fewer (hydrogen) filling stations • less efficient
5(b)(i)	• fractional distillation
5(b)(ii)	• cracking
5(c)	• Mr of $\text{CH}_3\text{CH}_2\text{COOCH}_3 = 88$ • mol of $\text{CH}_3\text{CH}_2\text{COOCH}_3 = 44 / 88 = 0.5$ • mol of H = $0.5 \times 8 = 4.0$ • no of H atoms = $4.0 \times 6.02 \times 10^{23} = 2.408 \times 10^{24}$
5(d)	• ethane • propene • ethanol • propanoic acid
5(e)(i)	• contain carbon and hydrogen atoms only
5(e)(ii)	• they have a carbon-carbon double bond
5(e)(iii)	• nickel
5(e)(iv)	• only one product is formed

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 49

14 marks

2024 Oct/Nov Paper 41 Question 6

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_w24_ms_41.pdf

6(a)	condensation
6(b)	proteins
6(c)	(3) - NH₂ group displayed - COOH group displayed - use of R
6(d)	- correct amide link between any two blocks showing all atoms and all bonds - identical orientation of inter-block amide links including terminal groups with correct orientation - continuation bonds on polymer
6(e)	- distance travelled by substance - distance travelled by solvent
6(f)(i)	pencil is insoluble (in solvent)
6(f)(ii)	locating agent
6(f)(iii)	same R_f value
6(f)(iv)	use a different solvent

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 50

9 marks

2024 Oct/Nov Paper 42 Question 1

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_w24_ms_42.pdf

1(a)	• G
1(b)	• H
1(c)	• B
1(d)	• D
1(e)	• E
1(f)	• B
1(g)	• A
1(h)	• H
1(i)	• G

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / M_r$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 51

11 marks

2024 Oct/Nov Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w24_ms_42.pdf

5(a)	• petroleum
5(b)	• high temperature • catalyst
5(c)(i)	• $\text{C}_{12}\text{H}_{26} \rightarrow 3\text{C}_4\text{H}_8 + \text{H}_2$ • C_4H_8 • correct equation
5(c)(ii)	• thermal decomposition
5(d)(i)	• poly(propene)
5(d)(ii)	• chain of six C atoms joined by single bonds in a chain • three correctly placed CH_3 groups • correct structure AND continuation bonds
5(d)(iii)	• addition

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 52

14 marks

2024 Oct/Nov Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w24_ms_42.pdf

6(a)(i)	• one amide linkage circled
6(a)(ii)	• two fully displayed carboxylic acid groups on the shaded box • two fully displayed amino groups on the unshaded box
6(a)(iii)	• water
6(a)(iv)	• protein(s)
6(a)(v)	• amino acids
6(a)(vi)	• displayed formula of 2-aminopropanoic acid (3) • displayed NH₂ or displayed COOH group present • displayed NH₂ and displayed COOH groups attached to same C atom • correct displayed structure of 2-aminopropanoic acid
6(b)(i)	• dicarboxylic acids • diols
6(b)(ii)	• any one correct fully displayed ester link between any two blocks • three fully displayed inter-block ester links with correct orientation • fully correct diagram showing two complete repeat units of polyester including continuation bonds

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 53

9 marks

2024 Oct/Nov Paper 43 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w24_ms_43.pdf

6(a)	• compound containing carbon and hydrogen only
6(b)(i)	• C ₁₂ H ₂₆
6(b)(ii)	• cracking
6(c)(i)	• contains at least one carbon - carbon double bond
6(c)(ii)	• C ₃ H ₆ Br ₂
6(d)(i)	• CH
6(d)(ii)	• addition
6(d)(iii)	• only two carbon atoms joined by a single bond and two additional bonds on each carbon • correct repeat unit including two extension bonds

Easy explanation / reminder

- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 54

14 marks

2025 Feb/March Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_m25_ms_42.pdf

6(a)	• (have a) general formula of C_nH_{2n} • (have a) $C=C$ functional group
6(b)	• compounds made of C and H only
6(c)	• $2C_3H_6 + 9O_2 \rightarrow 6CO_2 + 6H_2O$ • $CO_2 + H_2O$ as only products • correct equation
6(d)	• CH_2
6(e)	• but-2-ene
6(f)	• displayed formula of but-1-ene
6(g)(i)	• only one product is formed
6(g)(ii)	• orange to colourless
6(g)(iii)	• 2,3-dibromobutane
6(h)(i)	• displayed formula of ethanoic acid • February/March 2025
6(h)(ii)	• ethanoic acid
6(h)(iii)	• $CH_3CH=CHCH_3 + 2O_2 \rightarrow 2CH_3COOH$

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / M_r$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 55

10 marks

2025 May/June Paper 41 Question 1

Main topic: Topic 11: Organic chemistry

Cross-topic links: Topic 9: Metals, Topic 2: Atoms, elements and compounds, Topic 10: Chemistry of the environment

Answer source label: 0620_s25_ms_41.pdf

1(a)	• calcium oxide
1(b)	• graphite
1(c)	• ethanol
1(d)	• methane
1(e)	• bauxite
1(f)	• sulfur dioxide
1(g)	• propane
1(h)	• calcium oxide
1(i)	• oxygen
1(j)	• propene

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 56

18 marks

2025 May/June Paper 41 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s25_ms_41.pdf

5(a)	• concentration is no longer changing • rates of forward reaction and of the reverse reaction are equal
5(b)(i)	• rate decreases • yield of methanol decreases
5(b)(ii)	• no change(1) increases (1)
5(b)(iii)	• exothermic
5(b)(iv)	• copper • transition element
5(c)(i)	• any 2 from: • same or similar chemical properties or reactions • (same) general formula • (consecutive members) differ by CH₂ • same functional group • physical properties OR • example of a physical property • e.g. melting point / • boiling point / • volatility • vary in predictable manner or show trends or gradually change
5(c)(ii)	• C₅H₁₂O
5(d)(i)	• displayed formula of ester linkage, i.e. • H or C • completely correct displayed formula of methyl ethanoate or ethyl methanoate • i.e.
5(d)(ii)	• butan-1-ol • ethanoic acid
5(e)	• C 64.87 / 12 • H 13.51 / 1 • O 21.62 / 16 • OR • 5.41 13.51 1.35 • C₄H₁₀O

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 57

18 marks

2025 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_s25_ms_42.pdf

6(a)	homologous series
6(b)	No separate point listed.
6(c)	$\text{C}_2\text{H}_4 + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}$ $\text{CO}_2 + \text{H}_2\text{O}$ - correct equation
6(d)(i)	- decreases - decreases - decreases - no change
6(d)(ii)	- kinetic energy of particles increases - frequency of collisions between particles increases - higher percentage / proportion / fraction of collisions / particles have energy greater than / equal to activation energy
6(e)(i)	$\text{C}_5\text{H}_6\text{O}_4$
6(e)(ii)	it has a carbon-carbon bond which is not a single bond
6(e)(iii)	- one C - C single bond with continuation bonds - correct structure
6(e)(iv)	compound B has two carboxylic acid groups
6(e)(v)	(mol of NaOH reacting) = 0.2(00) - volume = $1000 \rightarrow 0.2(00) / 0.250 = 800 \text{ cm}^3$

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.

END OF ANSWER

Question 58

18 marks

2025 May/June Paper 43 Question 5

Main topic: Topic 11: Organic chemistry
Cross-topic links: Topic 6: Chemical reactions
Answer source label: 0620_s25_ms_43.pdf

5(a)	- the rate of forward reaction is equal to the rate of the reverse reaction - concentrations of reactants and products are no longer changing
5(b)(i)	- rate decreases - yield of ethanoic acid decreases
5(b)(ii)	- no - change - increases - increases
5(b)(iii)	- cobalt (it is a) transition element
5(c)(i)	methanoic acid
5(c)(ii)	C₄H₈O₂
5(d)(i)	- displayed formula of any ester linkage - displayed formula of methyl methanoate
5(d)(ii)	- butanoic acid - propan-1-ol
5(e)	- C 58.82 / 12 - H 9.80 / 1 - O 31.38 / 16 - OR 4.90:9.80:1.96 - C 4.90 / 1.96 - H 9.80 / 1.96 - O 1.96 / 1.96 - OR 2.5:5:1 - C₅H₁₀O₂

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 59

20 marks

2025 Oct/Nov Paper 41 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w25_ms_41.pdf

5(a)(i)	• $C_nH_{2n+1}OH$
5(a)(ii)	• (same) functional group
5(b)(i)	• 25-35 • 300 • yeast • acid • absence • of oxygen • 60 atm
5(b)(ii)	• $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ • $C_6H_{12}O_6$ as the only reactant on left • correct equation
5(b)(iii)	• $C_2H_4 + H_2O \rightarrow C_2H_5OH$ • both $C_2H_4 + H_2O$ only on left • correct equation
5(c)(i)	• $C_4H_{10}O_2$
5(c)(ii)	• No separate point listed.
5(d)(i)	• $C_2H_3O_2$
5(d)(ii)	• hydrogen
5(e)(i)	• any one correct fully displayed ester link between any two blocks • three fully displayed inter-block ester links with correct orientation • fully correct diagram showing two complete repeat units of polyester including continuation bonds
5(e)(ii)	• condensation (polymerisation)

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / M_r$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 60

12 marks

2025 Oct/Nov Paper 42 Question 5

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w25_ms_42.pdf

5(a)	• (have a) general formula of C_nH_{2n+2} • differ from one member to the next by a $-CH_2-$ unit
5(b)(i)	• increases
5(b)(ii)	• physical property • change as chain length increases
5(c)	• all carbon-carbon bonds are single (bonds)
5(d)(i)	• ultraviolet light
5(d)(ii)	• $CH_3CH_2CH_2Cl$ • 1-chloropropane • $CH_3CHClCH_3$ • 2-chloropropane
5(d)(iii)	• (compounds with) the same molecular formula but different structural formulae or structure

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 61

8 marks

2025 Oct/Nov Paper 42 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w25_ms_42.pdf

6(a)	• correctly circled ester link (including 2 C atoms and 2 O atoms)
6(b)	• ethyl butanoate
6(c)	• C ₃ H ₆ O
6(d)(i)	• butanoic acid • ethanol
6(d)(ii)	• acid
6(e)	• No separate point listed.
6(f)	• $2\text{C}_5\text{H}_{10}\text{O}_2 + 13\text{O}_2 \rightarrow 10\text{CO}_2 + 10\text{H}_2\text{O}$

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 62

17 marks

2025 Oct/Nov Paper 43 Question 6

Main topic: Topic 11: Organic chemistry
Answer source label: 0620_w25_ms_43.pdf

6(a)	• C 50(.00) / 12 • H 5.56 / 1 • O 44.44 / 16 • OR • 4.17:5.56:2.78 • fractions shown dividing all by smallest • OR • 1.5:2:1 • OR • 3:4:2 • C₃H₄O₂
6(b)	• C₄H₄O₄
6(c)(i)	• water
6(c)(ii)	• acid
6(c)(iii)	• propanoic acid • butan-1-ol
6(d)(i)	• addition
6(d)(ii)	• polymers are of indefinite length
6(d)(iii)	• 3 / three
6(d)(iv)	• C=C • displayed formula of but-2-ene • OR
6(e)(i)	• -NH₂ displayed • -COOH displayed • functional groups can be on either side
6(e)(ii)	• one circle around either amide linkage
6(e)(iii)	• proteins

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER